1. INVENTORY SYSTEM

THEORETICAL QUESTIONS

1. 2069 Q. No. 3

What is ABC inventory system? Explain material requirement planning for inventory management. [10]

2. 2069 (Old) Q. No. 6a

Explain the benefit of ABC inventory system and describe the major costs involved in holding inventory. [5]

3. 2068 Q.No. 6

What is material requirement planning? Discuss the material handling methods with example. [10]

4. 2067 Q.No. 4(Old)

Explain inventory model with shortage and lead time. Discuss how would you obtain EOQ an re-order point in both cases. [10

5. 2063 Q.No. 5

What is material requirement planning (MRP)? Discuss its merits and demerits. [10

6. 2063 Q.No. 9 (b)

State and explain various inventory management systems in deterministic conditions. [10

7. 2062 Q.No. 4

What do you understand by ABC classification in inventory management? Describe how ABC classification is applied in an organizations' inventory management system. [10]

8. 2061 Q.No. 5

Describe the ABC inventory planning system. What is the benefit of using inventory system?[10]

9. 2060 Q.No. 9(b)

What are the various types of inventory? State how would you obtain the optimum quantity level and reorder point in deterministic conditions. [10]

10. 2058 Q.No. 8 (b)

Describe the ABC inventory planning system. What is the benefit of using inventory system?[10]

NUMERICAL QUESTIONS

11. 2071 Q.No. 3

What is ABC inventory system? A company uses annually 50,000 units of an item costing Rs.1.20. It operates 250 days in a year with no lead time. Each order costs Rs.45 and inventory carrying cost is 15% of the item cost. Find the optimum order quantity. [10]

Ans: 5,000 units

12. 2070 Q.No. 10b

Apply the EOQ model to the following quantity discount situation in which D=500 units per year, CO=$40, the annual holding cost rate is 20%. What order quantity do you recommend?

Discount category	Order size	Discount (%)	Unit cost
1	0 to99	0	$10
2	100 or more	3	$9.70

Ans: EOQ = 100 units and 143 units; TVC = $5,282.84 and $5,130.7; More than 100 units should be order since it has minimum total cost is $5130.

13. 2069 Q. No. 7a

The demand of certain item per year is 100,000 units. The carrying cost per unit is Rs. 1.50 and ordering cost Rs. 100. Find the amount to be ordered at a time and also calculate reorder point. [5]

Ans: 3,652 units and 4,000 units (assuming L = 10 days)

14. 2069 (Old) Q. No. 6b

ABC Company requires 1000 units per month throughout the year at constant rate. If ordering cost are Rs. 250 per order, unit cost of the item is Rs. 25 and annual inventory cost are charged at 30%, determine the EOQ for the item. [5]

Ans: 894 units

15. 2068 Old Q.No. 6

Coca-cola Company has a soft drink product with constant annual demand rate of 3,000 cases. A case of soft drink costs the Co. Rs. 200. If ordering cost are Rs. 20 and inventory holding cost are charged at 25%, what is EOQ for this product? Also determine the cycle time (in days). [10]

Ans: 49 cases and 5 days (assuming 300 working days in a year)

16. 2065 Q.No. 6

Explain the ABC analysis as inventory management technique. An aircraft company uses rivets at an approximate rate of 2,500 kg per year. The rivets costs Rs. 30 per kg and the company personnel estimates that it costs Rs. 1.30 to place on order and the inventory carrying costs is 10% per year. How frequently should orders for rivets be placed and what quantities should be ordered? [10]

Ans: EOQ= 47 kg; Optimum number of order = 54 times; Time between order (cycle)= 6.67 days

17. 2064 Q.No. 9 b

Each year the 'Redstone Company' purchases 20,000 of an item that costs Rs. 16 per unit. The cost of placing an order is Rs. 12 and the cost to hold the item for one year is 24% of the unit cost. Determine the economic order quantity (EOQ) and the average inventory level, assuming that the minimum inventory level is zero. [10]

Ans: EOQ = 354 units and 177 units

18. 2062 Q.No. 6

A manufacturing organization experienced constant annual demand of 10,000 unit of an item which cost Rs. 40 per unit. If the order cost is Rs. 20 per order and the holding cost is 25% of the item costs determine the economic order quantity and the reorder level with no lead-time known. [10]

Ans: EOQ = 200 units, ROL = 40 L units

19. 2061 Q.No. 9 (b) (i)

A manufacturer has to supply his customer with 600 units of his product per year. Shortages are not allowed and the storage cost amounts to Rs. 0.6 per unit per year. The set up cost per run is Rs. 80. Find the optimum run size and the minimum average yearly cost. [5]

Ans: 400 units and Rs. 240

20. 2060 Q.No. 6

Nepal soft drink Co. has a soft drink product which has a constant annual demand rate of 3000 cases and cost Rs. 200/case.if ordering cost are Rs. 20 and inventory holding cost are charged at 25%, what is the EOQ for this product? Also determine the cycle time (in days). [10]

Ans: EOQ = 49 cases and 6 days

21. 2059 Q.No. 3

What are the main objectives of holding inventory in an organization? What are the major costs involved in holding inventory? The ABC requires 100 units per months though out the year at constant rate. If ordering cost are Rs. 250 per order, unit cost of the item is Rs. 25 and annual inventory holding cost are charged at 30%, their determine the EOQ for the item. [10]

Ans: 1095.45 units